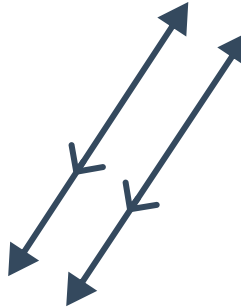


Parallel lines and transversals



- 1
- a Parallel lines are indicated using arrows or 'chevrons'. The number of arrows connects which lines are parallel.
 - b Example answer:



- 2
- a Corresponding angles
 - b Alternate angles
 - c Cointerior angles

- 3
- a Equal
 - b Supplementary
 - c Equal

- 4
- a
 - i $\angle EPQ$
 - ii $\angle APQ$
 - iii $\angle APF$
 - b
 - i $\angle CQP$
 - ii $\angle EQP$
 - iii $\angle DQE$

- 5
- a The diagram contains parallel lines. The marked angles are supplementary and cointerior so we have a transversal of parallel lines.
 - b The diagram does not contain parallel lines. The marked angles are not supplementary so they are not cointerior. We therefore do not have a transversal of parallel lines.
 - c The diagram contains parallel lines. The marked angles are equal and corresponding so we have a transversal of parallel lines.
 - d The diagram does not contain parallel lines. The marked angles are not equal so they cannot be corresponding. We therefore do not have a transversal of parallel lines.
 - e The diagram contains parallel lines. The marked angles are equal and alternating so we have a transversal of parallel lines.
 - f The diagram contains parallel lines. The marked angles are not equal so they cannot be alternating. We therefore do not have a transversal of parallel lines.

- g** The diagram does not contain parallel lines. The marked angles are not equal so they cannot be corresponding. We therefore do not have a transversal of parallel lines.
- h** The diagram contains parallel lines. The marked angles are supplementary and cointerior so we have a transversal of parallel lines.



6

- a** $x = 125$ **b** $x = 75$ **c** $x = 55$ **d** $x = 52$
e $x = 107$ **f** $x = 96$

7

- a** They are alternate angles, so they are also equal.
b They form a straight angle, so they are also supplementary.
c 96°

8

- a** They are vertically opposite angles, so they are also equal.
b They are cointerior angles, so they are also supplementary.
c 73°

9

- a** They are cointerior angles, so they are also supplementary.
b They form a straight angle, so they are also supplementary.
c 97°

10

- a** $x = 33$ **b** $y = 57$